

DIRECTIONS *for questions 1 to 5:* Answer these questions on the basis of the information given below.

The currency of the country Banama is Rupas (Rp). There are exactly five denominations in which the currency is available – Rp 2, Rp 3, Rp 7, Rp 17 and Rp 19.

Each of five persons, A through E, had notes of exactly one denomination with him and no two persons among the five had notes of the same denomination. On a particular day, the five persons visited a shop and each of them purchased a different product. The amounts that A through E paid the shopkeeper were Rp 102, Rp 357, Rp 399, Rp 238 and Rp 147, in that order. Further, the number of notes that each person gave the shopkeeper was distinct. However, the amount that each person paid the shopkeeper was more than the price of the product that he purchased. Therefore, the shopkeeper returned the excess amounts to each of A through E in the form of 2 notes, 4 notes, 1 note, 2 notes and 1 note respectively, but in denomination(s) different from that in which the person had paid the shopkeeper.

Q1. DIRECTIONS *for question 1:* Type in your answer in the input box provided below the question.

What is the total number of notes that the five persons paid the shopkeeper?

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Q2. DIRECTIONS for questions 2 to 5: Select the correct alternative from the given choices.

Which of the following cannot be the price (in Rp) of the product that A purchased?

☐ a) 76

☐ b) 92

☐ c) 80

☐ d) 82

Q3. DIRECTIONS *for questions 2 to 5:* Select the correct alternative from the given choices.
Which of the following statements is/are definitely true?

- I. The price of the product that A purchased is an even number.
- II. The price of the product that B purchased is an even number.
- III. The price of the product that C purchased is an odd number.
- IV. The price of the product that D purchased is an even number.

- ☐ a) **Only I and IV**
- ☐ b) **Only IV**
- ☐ c) **Only I, II and III**
- ☐ d) **Only II and IV**

Q4. DIRECTIONS *for questions 2 to 5:* Select the correct alternative from the given choices.

Which of the following statements is sufficient to determine the price of the product that E purchased?

- ☐ a) **The price of the product is not a multiple of 10.**
- ☐ b) **The price of the product is a multiple of 10.**
- ☐ c) **The price of the product is not a multiple of 5.**
- ☐ d) None of the above

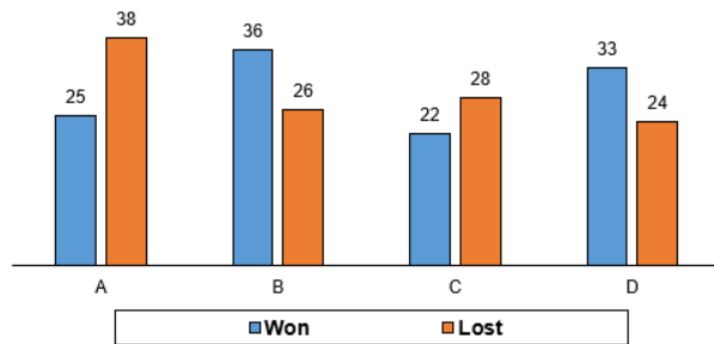
Q5. DIRECTIONS *for questions 2 to 5:* Select the correct alternative from the given choices.

What is the denomination of the notes that B had?

- ☐ a) **Rp 3**
- ☐ b) **Rp 7**
- ☐ c) **Rp 17**
- ☐ d) **Rp 19**

DIRECTIONS for questions 6 to 10: Answer the questions on the basis of the information given below.

The bar graph below provides the number of wars won and lost by four countries – A through D – during the 18th and the 19th centuries. It is known that, during the 18th and 19th centuries, the four countries waged wars only against each other.



Q6. DIRECTIONS for question 6: Select the correct alternative from the given choices.

If D won 16 wars against each of B and C, what is the maximum number of wars that B won against D?

- ☐ a) 21
- ☐ b) 22
- ☐ c) 23
- ☐ d) 24

Q7. DIRECTIONS *for question 7:* Type your answer in the input box provided below the question.

If B won an equal number of wars against each of A, C and D, what is the maximum number of wars that D could have won against A?

Q8. DIRECTIONS *for question 8:* Select the correct alternative from the given choices.

Among the four countries, it so happened that a pair of countries (X, Y) were such that all the wars that X waged were only against Y.

How many possibilities exist for (X, Y)?

- ☐ a) 1
- ☐ b) 2
- ☐ c) 3
- ☐ d) 4

Q9. DIRECTIONS *for question 9:* Type your answer in the input box provided below the question.

Among the four countries, it so happened that a pair of countries (X, Y) were such that all the wars that X waged were only against Y.

What is the maximum number of wars that D won against B?

Q10. DIRECTIONS *for question 10:* Select the correct alternative from the given choices.

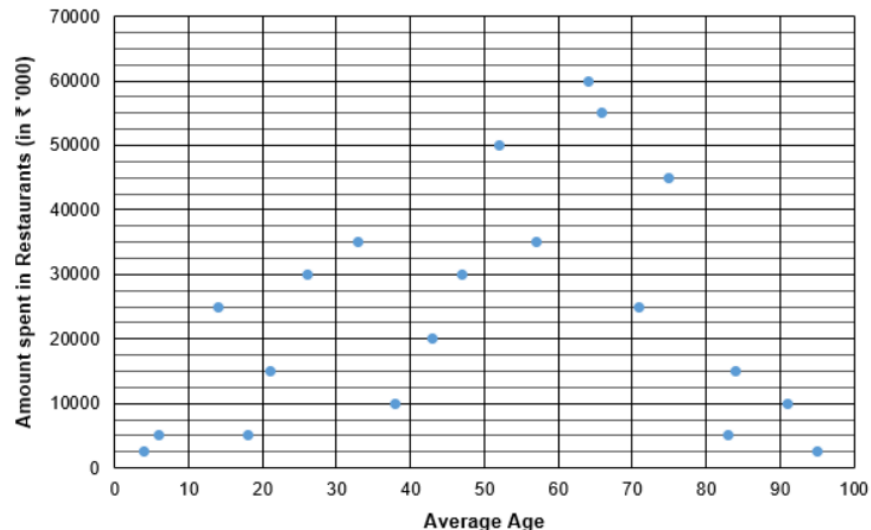
If D won exactly 20 wars against A and lost exactly 20 wars against A, what is the minimum number of wars that B won against C?

- ☐ a) 0
- ☐ b) 10
- ☐ c) 14
- ☐ d) 18

DIRECTIONS for questions 11 to 15: Answer these questions on the basis of the information given below.

Kalyan, as a part of his college assignment, was studying the total amount spent in restaurants by the populations of two countries, Country A and Country B, across different age groups. He divided the entire population of each country into ten age groups: 0-10 years, 11-20 years, 21-30 years and so on up to 91-100 years.

After collecting the data, he plotted a scatter chart, given below, with the horizontal axis representing the average age (in years) of the persons in each age group and the vertical axis representing the total amount spent in restaurants (in ₹ '000) by all the persons in that group. In his hurry to complete the assignment on time, he forgot to label the points in the graph. However, he knew that, for any age group, the average age of all the persons in that age group in Country A was greater than that in Country B.



Q11. DIRECTIONS for question 11: Type your answer in the input box provided below the question.

What is the total amount spent (in ₹ '000) in restaurants by all the persons of Country A whose age is at least 31 years?

Q12. DIRECTIONS *for questions 12 to 14:* Select the correct alternative from the given choices.

How many of the following can be the total amount (in ₹ '000) spent by all the persons of Country B whose age is between 36 years and 65 years?

I. 65000

II. 95000

III. 150000

IV. 170000

V. 195000

☐ a) 1

☐ b) 2

☐ c) 3

☐ d) 4

Q13. DIRECTIONS *for questions 12 to 14:* Select the correct alternative from the given choices.

For which age group is the ratio of the total amount spent in restaurants by all the people of that age group in Country A to that in Country B the highest?

- ☐ a) 0-10 years
- ☐ b) 11-20 years
- ☐ c) 71-80 years
- ☐ d) 81-90 years

Q14. DIRECTIONS *for questions 12 to 14:* Select the correct alternative from the given choices.

If, in each age group, there were an equal number of persons in Country A and in Country B, in how many age groups was the amount spent in restaurants per person in Country B at least 25% more than that in Country A?

☐ a) 5

☐ b) 4

☐ c) 3

☐ d) 0

Q15. DIRECTIONS *for question 15:* Type your answer in the input box provided below the question.

What is the total amount (in ₹ '000) spent in restaurants by all the persons of Country B whose age is at most 40 years?

DIRECTIONS for questions 16 to 20: Answer the questions on the basis of the information given below.

Five friends – Faraday, Gauss, Hertz, Joule and Kelvin – purchased five cats, each of a different breed among Abyssinian, Persian, Ragdoll, Siamese, and Sphynx. Each friend purchased his cat at different time. The following information is known about when the five friends purchased their cats:

- i. Kelvin purchased his cat, which was not Siamese, before Gauss purchased his cat, which was not Persian.
- ii. Hertz purchased his cat, which was a Ragdoll, before Faraday.
- iii. Joule purchased his cat, which was neither Siamese nor Abyssinian, after Hertz but before Kelvin.
- iv. Faraday purchased his cat, which was not any of Persian, Siamese or Sphynx, after Gauss.

Q16. DIRECTIONS for questions 16 to 18: Select the correct alternative from the given choices.

Which breed of cat did Faraday purchase?

- ☒ a) **Abyssinian** ✓ Your answer is correct
- ☐ b) **Persian**
- ☐ c) **Spynx**
- ☐ d) **Siamese**

Q17. DIRECTIONS *for questions 16 to 18:* Select the correct alternative from the given choices.

Who purchased a cat immediately after a Siamese cat was purchased?

☐ a) Kelvin

☐ b) Hertz

☐ c) Joule

☒ d) Faraday ✓ Your answer is correct

Q18. DIRECTIONS *for questions 16 to 18:* Select the correct alternative from the given choices.

If all the cats were of the same exact age when they were purchased, which of the following is the youngest?

- ☐ a) **The Siamese cat**
- ☐ b) **The Ragdoll cat**
- ☒ c) **The Abyssinian cat** ✓ **Your answer is correct**
- ☐ d) **The Persian cat**

Q19. DIRECTIONS *for question 19:* Type in your answer in the input box provided below the question.

In how many ways could the five friends have purchased the different breeds of cats?

Q20. DIRECTIONS *for question 20:* Select the correct alternative from the given choices.

Who was the first person to purchase a cat?

- ☐ a) Kelvin
- ☐ b) Faraday
- ☒ c) Hertz ✓ Your answer is correct
- ☐ d) Joule